

NFS-TASK

POC 8 — Two Pods Sharing One PVC

Question / Task

POC 8 — Two Pods Sharing One PVC

Reuse (or create another) PVC.

Create pod-a and pod-b (both toolbox image), mounting the same PVC at `/data` .

```
pod-a: kubectl exec pod-a -- sh -c 'echo shared-data > /data/test.txt'
```

```
pod-b: kubectl exec pod-b -- cat /data/test.txt
```

Expected result: If both pods land on the same node and the PV is ReadWriteOnce, pod-b successfully reads `shared-data` — RWO restricts the volume to a single node, not a single pod, so co-located pods can share it. If the scheduler places pod-b on a different node, the mount fails or the pod stays stuck in `ContainerCreating` , since an RWO volume can't attach to two nodes at once. A ReadWriteMany PV (e.g. NFS-backed) would succeed regardless of node placement.

Solution / Execution

```
helm version
helm repo add csi-driver-nfs https://raw.githubusercontent.com/kubernetes-
csi/csi-driver-nfs/master/charts
helm repo list
helm repo update
helm install csi-driver-nfs csi-driver-nfs/csi-driver-nfs --namespace kube-
system
```

```
kubectl get pods -n kube-system | grep nfs
kubectl --namespace=kube-system get pods --
selector="app.kubernetes.io/instance=csi-driver-nfs" --watch
curl -skSL https://raw.githubusercontent.com/kubernetes-csi/csi-driver-
nfs/v4.13.2/deploy/install-driver.sh | bash -s v4.13.2 -- (optiona for helm
chart)
```

```
payal_kharat@LAPTOP-# cat nfs-servipayal_kharat@LAPTOP-# cat nfs-server.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nfs-server
  namespace: revision-1
spec:
  replicas: 1
  selector:
    matchLabels:
      app: nfs-server
  template:
    metadata:
      labels:
        app: nfs-server
    spec:
```

```
containers:
  - name: nfs-server
    image: itsthenetwork/nfs-server-alpine:latest
    securityContext:
      privileged: true
    env:
      - name: SHARED_DIRECTORY
        value: /exports
    ports:
      - name: nfs
        containerPort: 2049
        protocol: TCP
    volumeMounts:
      - name: exports
        mountPath: /exports
volumes:
  - name: exports
    emptyDir: {}
-----
payal_kharat@LAPTOP-# kubectl apply -f nfs-server.yaml -n revision-1
deployment.apps/nfs-server created
payal_kharat@LAPTOP-# kubectl get pods -n revision-1
NAME                                READY   STATUS    RESTARTS   AGE
nfs-server-86694c4b7c-2zqkp        1/1     Running   0           3m2s
-----
payal_kharat@LAPTOP-# cat service.yaml
apiVersion: v1
kind: Service
metadata:
  name: nfs-server
  namespace: revision-1
spec:
  type: ClusterIP
  selector:
    app: nfs-server
  ports:
    - name: nfs
      port: 2049
      targetPort: 2049
      protocol: TCP
-----
payal_kharat@LAPTOP-# kubectl apply -f nfs-service.yaml -n revision-1
service/nfs-server created
payal_kharat@LAPTOP-# kubectl get svc -n revision-1
NAME            TYPE           CLUSTER-IP      EXTERNAL-IP  PORT(S)    AGE
nfs-server      ClusterIP      10.110.240.141  <none>       2049/TCP   2m37s
-----
payal_kharat@LAPTOP-# cat nfs-sc.yaml
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
  name: nfs-csi
provisioner: nfs.csi.k8s.io
parameters:
  server: nfs-server.revision-1.svc.cluster.local
  share: /
  onDelete: delete
mountOptions:
  - nfsvers=4.1
-----
```

```
payal_kharat@LAPTOP-# kubectl apply -f nfs-sc.yaml -n revision-1
storageclass.storage.k8s.io/nfs-csi created
payal_kharat@LAPTOP-V1NV0QFT:Revision-1$ kubectl get sc -n revision-1
NAME                                PROVISIONER                                RECLAIMPOLICY
VOLUMEBINDINGMODE  ALLOWVOLUMEEXPANSION  AGE
nfs-csi            nfs.csi.k8s.io       Delete          Immediate
false              2m23s
standard (default) k8s.io/minikube-hostpath Delete          Immediate
false              19m
-----
-
payal_kharat@LAPTOP-# cat pvc.yaml
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: shared-data
  namespace: revision-1
spec:
  accessModes:
    - ReadWriteMany
  storageClassName: nfs-csi
  resources:
    requests:
      storage: 1Gi
-----

payal_kharat@LAPTOP-# kubectl apply -f pvc.yaml -n revision-1
persistentvolumeclaim/shared-data created
payal_kharat@LAPTOP-V1NV0QFT:Revision-1$ kubectl get pvc -n revision-1
NAME          STATUS    VOLUME                                     CAPACITY
ACCESS MODES  STORAGECLASS  VOLUMEATTRIBUTESCLASS  AGE
shared-data   Bound        pvc-02fb4541-7972-4309-ac29-410fbe3ab99b  1Gi
RWX           nfs-csi      <unset>                  2m11s
payal_kharat@LAPTOP-V1NV0QFT:Revision-1$ kubectl get pods -n revision-1
-----

payal_kharat@LAPTOP-# cat nfs-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nfs-deployment
  namespace: revision-1
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nfs-deployment
  template:
    metadata:
      labels:
        app: nfs-deployment
    spec:
      containers:
        - name: test
          image: busybox:1.36
          command:
            - sh
            - -c
            - |
              echo "Pod started: $(hostname)"
              sleep 3600
          volumeMounts:
```

```
      - name: shared
        mountPath: /shared
volumes:
  - name: shared
    persistentVolumeClaim:
      claimName: shared-data
```

```
payal_kharat@LAPTOP-# kubectl apply -f nfs-deployment.yaml -n revision-1
deployment.apps/nfs-deployment created
payal_kharat@LAPTOP-V1NV0QFT:Revision-1$ kubectl get deployments -n revision-1
```

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
nfs-deployment	2/2	2	2	2m26s
nfs-server	1/1	1	1	4m41s

```
payal_kharat@LAPTOP-# kubectl get pods -n revision-1
```

NAME	READY	STATUS	RESTARTS	AGE
nfs-deployment-85fdb8d8c4-85grp	1/1	Running	0	2m19s
nfs-deployment-85fdb8d8c4-phwt9	1/1	Running	0	2m19s
nfs-server-86694c4b7c-2zqkp	1/1	Running	0	4m34s
